

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element H Initial Template

How to use this template

Guidance before filling out the table on the next page:

- Enter *all* of the high-priority design requirements you listed in Element C into the table below.
- Determine an appropriate testing method for each requirement. For example, the test may involve measurement, inspection, physical actions, or mathematical modeling.
- As you plan, use your scientific method best practices, e.g. creating experimental controls, defining independent and dependent variables, only changing one variable at a time and changing it in increments to find a trend, state your prediction, etc.
- Your testing plan for a given requirement must include enough detail so that a person who is not on the design team could fully replicate the test. Include photos and/or diagrams to add clarity. Each test should include the following details:
 - how the test will be set up
 - what needs to happen during the test
 - test cases (if applicable)
 - number of trials of the test expected (aim for 7 or more trials)
 - the expected range of results that could be recorded, including units
 - the criteria for passing or failing the associated design requirement, including any permissible range for error and/or minimal thresholds
 - any goals your team has set beyond the passing result

Element H: Prototype Testing and Data Collection Plan

Testing Details (H1, H2, H3, H4)

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1 ex	The prototype must remain intact when held upside down.	1) Hold the prototype in a hand using the grip in this photo. <insert grip photo> 2) Take 3 seconds to slowly rotate the prototype around the indicated axis into an upside down position with a smooth motion. <insert photo with annotation of axes of rotation> 3) Note any observations about prototype integrity. 4) Take 3 seconds to rotate the prototype in reverse, back to its original position, using a smooth motion	a. Rotate around x-axis (10 trials) b. Rotate around y-axis (10 trials) c. Rotate around z-axis (10 trials)	Parts of the prototype may: – remain unchanged – deform – shift position – tear – detach	Fail: one or more parts of the prototype tears or detaches Pass: zero parts of the prototype tear or detach Goal: all parts of the prototype remain unchanged when the prototype is returned to its original position Pass
2 ex	mass of prototype < 3kg	1) Tare a digital scale. 2) Place the prototype securely on the scale. 3) Wait for the reading on the scale to stabilize before recording the value.	test cases: N/A # trials: 10	0kg<reading<10kg	Pass: mean of the trials ≥ 3kg Fail: mean of the trials < 3kg Goal: lighter is better for maneuverability Pass was ~2kg
1	Must collect MP	1) Put beads or small bits of plastic in a	We get 3 chances	Collects half or	Pass:

	and filter out water	tub (floating and at the bottom of the tub) 2) sent a timer for 5 min 3) put the collector in the water and see how much you can collect in that time (move it by hand) 4) if it collects half or more than it passes, and it filters out most of the water	to do this In each trail you can scoop as much as you want, it just needs to be within the time	more than half of MP and filters out the water	Collects 50% + of the MP No excess water Fail: collects less than 50% of MP Does not filter out the water
2	Can the ROV move in the water	1) place the ROV in the tub 2) 3 min timer 3) Can it move smoothly in all directions (Up, Down, Left, Right, Forward, Backwards)	We have 5 attempts to achieve this	It can move smoothly in all directions we want it to go in	Pass: It moves how we want it to and can go smoothly in all directions Fail: Does not move where we want it to move Not able to go in one or more directions It sinks and does not move
3	After every test the robot maintains its function	1) Put the ROV in water and move it a 2) observe and record its movement 3) revolve from water 4) place it back in water and move it around 5) observe and record its movement 6) compare the two observations	We have 3 chances for them to be comparable	The tests show that the movement stayed the same and it did not lose function	Pass: The movement stays the same Fail: The ROV loses movement or stops moving
4	Do the controls	1) put the ROV on a table	We have 3 chances	1 goes to one	Pass:

	make sense to the operator	2) test all the controls 3) determine if they make sense	to have them make sense	2 goes to two 3 goes to three The ups and downs make it go the same ways	The controls are intuitive Fail: The controls are all over the place
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Feedback From Field Experts (H5)

Guidance to students about this section of Element H.

- a) "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.
- b) You must provide evidence in this section that your testing plan has been reviewed by 2-3 field experts.
- c) Do not incorporate their feedback in the table above. Rather, make a copy of the table in the section above, paste it into the section below, and make all revisions only in the table below. This provides evidence that you incorporated the feedback and shows your testing plan's progression.

Revised Testing Details That Incorporate Field Expert Feedback (H5)

After talking to Mary we had a few suggestions but after having a chance to explain our situation a little more we decided on keeping all goals the same.

The one thing that was changed is that sampling is prioritized over movement, since theoretically that's the only thing that our clients would care about.